



AI Inventorship, Legal-Philosophical Justice, and Harmonising Uzbekistan’s Patent Law with WIPO and EPO Standards

Pyali Chatterjee¹, Nodirbek Inoyatov^{2*}, Eshonqulov Javokhir³, Dadakhon Khudoynazarov⁴, Nargiza Sulaymonova⁵

¹ Faculty of Law, ICFAI University, Raipur, Chhattisgarh, 490042, India
^{2, 3, 4, 5} Tashkent State University of Law, Tashkent, 100000, Uzbekistan
* Corresponding author: nodirbekinoyatov8@gmail.com

Article	Abstract
Keywords: Accountability; Artificial Intelligence; Comparative Law; Inventorship; Patentability. Article History Received: Nov 16, 2025; Reviewed: Dec 20, 2025; Accepted: Sep 12, 2026; Published: Sep 18, 2026.	<i>The paper examines how these jurisdictions define inventorship, allocate responsibility, and regulate the interaction between human and machine creativity within the current intellectual property protection system. Whether Artificial Intelligence (AI) systems can qualify as inventors under national and international patent law has become as one of the most contested issues in contemporary intellectual property doctrine. The findings indicate that WIPO advocates for a neutral “AI-assisted inventorship” model, emphasising global harmonisation, whereas the EPO follows a human-centred approach that allows only natural persons to be considered inventors. This article argues that the core regulatory challenge is not recognising AI as an inventor, but allocating legal responsibility for AI-assisted inventive outputs. Uzbekistan’s patent law reserves inventorship exclusively for natural persons, thereby excluding AI from independent recognition. Recent AI governance instruments in Uzbekistan signal incremental alignment with WIPO standards (Presidential Resolution RP-358, 2024). These instruments do not alter Uzbekistan’s foundational patent doctrine. The study concludes that AI need not be recognised as an inventor to comply with WIPO and EPO standards. Instead, three mechanisms can close the gap without any legislative overhaul. These include mandatory disclosures of AI involvement in patent applications, mediated attribution rules that allocate responsibility among developers and applicants; and deepened institutional cooperation with the EPO. Together they offer Uzbekistan a path to meaningful alignment with international patent standards. The article further argues that the patentability of AI-generated inventions implicates questions of attribution justice, access equity, and accountability fairness that reform frameworks must address alongside doctrinal harmonisation.</i>



Copyright ©2026 by Author(s); This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License. All writings published in this journal are personal views of the authors and do not represent the views of this journal and the author's affiliated institutions.

INTRODUCTION

The patent office first confronted AI inventorship in 2018, when applications for AI-generated innovations emerged (Bayındır & Danaher, 2025). Patent law has addressed this matter; however, it does not apply here. The conception theory of inventorship, as established by *Burroughs Wellcome Co. v. Barr Labs.* (1994), stipulates that the inventor must possess a complete and definite invention (Abbott, 2020). In other words, the inventor must go through a process of conceptualising that only humans are legally entitled to perform (Błaszczuk, 2024). The issues of patenting and allocating responsibility for AI-generated inventions present a conceptual conundrum for patent law.

DABUS (Device for the Autonomous Bootstrapping of Unified Sentience), an artificial neural network created by Stephen Thaler, has generated two AI-driven innovations (Matulionyte, 2025). These innovations included a fractal-geometry food container and an emergency light beacon. In an unprecedented move, Thaler identified “the machine” as the inventor and applied to the United States Patent and Trademark Office, the European Patent Office, and the United Kingdom Intellectual Property Office with the assistance of the agent (Oriakhogba, 2021). In each application, he indicated that he is the owner and applicant of the invention. Each patent office had to determine whether an AI system could be considered an inventor. The responses showed that patents awarded to machines are indeed a conundrum for international patent law.

At the same time, the responses of South Africa, Australia, and Switzerland differ. For example, South Africa was the first country to accept an AI-invented invention, registering it in 2021 (Sabel Nastasya & Santoso, 2025). As a result, responses varied across countries, raising concerns that applicants would exploit these discrepancies to take advantage of the relative leniency of particular jurisdictions (Drahos, 2021). This divergence prompted researchers to examine the issue's theoretical premises and practical ramifications (Novelli et al., 2024).

Bundle theory is a theoretical approach to analyse the issue. This theory suggests that legal personhood is a set of incidents that must be acquired or that can be bestowed upon an entity. These include rights, duties, standing, and liability (Li et al., 2025). Notably, bundle theory allows assessment of the incidents of inventorship and whether they can be granted to AI (Aboy et al., 2025).

The issue of inventorship and liability extends beyond patent law. For example, a constitutional analysis of US patent law and AI demonstrates that vague eligibility criteria prompt inconsistencies and arbitrariness across jurisdictions (AllahRakha,

2025a). The conversations involving the World Intellectual Property Organization (WIPO) on IP and AI from 2019-2023 emphasise the importance of human responsibility in AI-driven innovation. Specifically, the discussions show that humans' responsibility for AI inventions is inalienable, despite their relative insignificance in the inventive process (Al-Busaidi et al., 2024). The reviewed articles demonstrate that the focus of the debate should turn from the obvious ability of AI to generate inventions to the manner in which their creator should bear responsibility for AI-generated inventions (AGIs) (Oriakhogba, 2021). The disclosure-based accountability model is a viable option for patent law, as it recognises AI's role in innovation while preventing the inalienability of legal personhood.

Notably, the WIPO conversations, which took place over three years, have resulted in broader recognition of the principles of human responsibility for AGIs (Ahmed Mohamed et al., 2025). An AI-assisted invention involves a human inventor. By contrast, under Uzbekistan's current law, an AGI has no eligible inventor (Yohanna & Suleiman, 2025). This article addresses the following research questions: first, how have WIPO and the EPO acknowledged the challenges AGIs pose within the confines of human inventorship? Second, can Uzbekistan's current IP law accommodate AI inventorship in any capacity? If the response to the latter question is negative, what changes can be implemented to harmonise Uzbekistan's IP law with that of WIPO and the EPO without interfering with Uzbekistan's current patent doctrines?

This article makes three distinct contributions to the scholarly literature on AI and patent law. First, it develops an integrated comparative analysis of WIPO standards, EPO jurisprudence, and Uzbekistan's patent law, which has not been undertaken previously. Second, the study uses bundle theory and Hohfeldian analysis to assess the reform options available to Uzbekistan (Balganesh et al., 2022). The study outlines reform recommendations for Uzbekistan that take into account existing national circumstances and the opportunities presented by Uzbekistan's membership in the EAPO. The study underscores the underutilised potential of Uzbekistan's EAPO membership in addressing the challenges posed by AGIs for patent law. It extends the disclosure-based accountability model to AI-assisted green-technology patents, tying Uzbekistan's patent-law reform to its climate and technology-transfer obligations under the Paris Agreement.

METHOD

The study employs three legal methods, encompassing doctrinal, comparative, and normative analysis. The doctrinal analysis identifies what each legal framework currently offers in terms of legal definitions. The comparative analysis then compares the WIPO, EPO, and Uzbekistan frameworks to identify similarities and gaps. Finally, the normative analysis evaluates the findings against the criteria of accountability, transparency, and innovation-policy coherence to inform the recommendations for regulatory reform.

The study uses primary and secondary sources. The primary sources include WIPO-administered treaties, notably the Agreement on Trade-Related Aspects of Intellectual Property Rights, the Paris Convention for the Protection of Industrial Property, WIPO conversations on intellectual property and artificial intelligence (IP/AI) 2019-2023, and policy notes. In the case of the European Union (EU), the study will consider the European Patent Convention Article 81, the Legal Board of Appeal of the EPO, Decision J 8/20 and J 9/20, EPO Guidelines for Examination of AI inventions, and the EU AI Act 2024. As for Uzbekistan, the study will focus on the Law on Inventions, Utility Models and Industrial Designs, and Presidential Resolution RP-358 (2024). The secondary sources include articles published in Scopus- and WoS-indexed journals. WIPO, the EPO, and Uzbekistan were selected because they represent diverse jurisdictions with varying yet overlapping regulatory solutions. WIPO treaties offer a global perspective on regulating AI-generated inventions, while the EPO represents a well-established patent office with extensive jurisprudence on contentious and emerging technologies. Uzbekistan is a developing jurisdiction with limited regulations governing AI innovations.

This study uses bundle theory as its theoretical framework. This theory assumes that bundle items, including rights, duties, standing, and liability, among others, are independent and can be separated (Kurki, 2019). Bundle theory here identifies the specific bundle items that can be used to define an inventor. Beyond identifying specific items, the theory is also used to argue whether AI can be regarded as an inventor based on whether it can demonstrate that it has bundle items. The theory helps highlight the differences between the inventor's ability to perform or not perform specific actions, notably, the ability to contribute to an invention and the ability to be assigned as an inventor. In other words, the theory helps separate the functional question of whether AI can perform an invention from the normative question of whether it can be regarded as an inventor. This study focuses on legal analysis rather than human behaviour. However, this study lacks statistical data, and future studies can supplement this research by analysing the practices of the Uzbekistan IP Department in handling patents.

RESULTS AND DISCUSSION

For example, in the pharmaceutical industry, if researchers direct AI to search for compounds within certain parameters, this is an AI-assisted discovery with a human inventor. However, if AI independently identifies, validates and characterises a drug candidate without human involvement at the level of core inventive decisions, it is an AI-generated invention. In this case, Uzbekistan legislation does not allow formal recognition of any inventor because no human is involved at the stage of invention (Kim et al., 2022).

Justice and Legal-Philosophical Dimensions of AI Inventorship

The problems related to the patentability of AGIs go beyond doctrinal differences and enter the realm of legal philosophy and the just allocation of costs and benefits. The review revealed three essential justice-related challenges of AGI inventorship. A conflict exists between the principle of adequate reward for invention and the principle of full disclosure of information about the invention (Kim, 2020). If an AI-assisted invention falls under the second category, then the situation emerges in which a human who did not contribute to the inventive step must be indicated as the inventor (DiMatteo et al., 2022). Under current legislation, a human who did not participate in creating the invention receives rights to it, while AI, as the actual creator, receives no benefits from the patent (Sabbagh, 2001). Thus, a human who is indicated as the inventor receives disproportionate rewards in comparison with the contribution to the invention, which falls under the definition of attribution injustice (Rodrigues, 2020).

When an invention is created by AI, the legal entity or entrepreneur who owns the AI is typically held most responsible for the resulting problems. If a human has no full visibility of the processes occurring in AI, then his or her liability for the consequences of the invention is disproportionate to the level of control over AI. On the contrary, responsibility is a prerequisite for exercising any freedoms and their corresponding rights. This formulation implies that legal and ethical principles of justice should also consider cost-benefit balance and risk allocation. A just solution would be for companies to bear greater responsibility for AI activities by implementing a transparent mechanism to establish the inventor to ensure the protection of the rights of all parties involved (Matsieli & Mutula, 2025).

In the context of the legal-philosophical foundations, the current solution to the problem of the right to patent an AI invention fits within a system of stable justice by supporting natural-person-based relations and the associated rights and duties. This approach emphasises a philosophical position entirely consistent with the principles of positivist legal theory, since it prioritises the legal method over ethical norms. At the same time, this approach also raises several challenges related to justice (Mahlmann, 2023). The exclusion of AI from the status of the creator is explained by its inability to meet the requirements of responsibility, awareness, and accountability, which are essential for a legal person. This formula does not contradict the principle of justice, but the analysis shows that it implies additional challenges in applying the law. In particular, when the rights of protection and the duties of accountability fall on humans who lack knowledge of the processes performed by AI, the principle of justice in terms of benefits and harms begins to operate unfairly. In this regard, a just solution would be for the human actors behind the invention to explicitly denote AI as the creator.

Legal Frameworks for AGIs

Comparing the legal frameworks on AGIs yields three conclusions. First, according to Kwakwa (2019), none of the treaties under the WIPO (the WIPO Performances and Phonograms Treaty, Paris Convention, TRIPS) allows for expanding the notion of who can be an inventor beyond humans (Kwakwa, 2019). Therefore, changing this paradigm would require a treaty amendment that no country is willing to accept. Secondly, at the policy level, WIPO negotiations have converged on the notion of “AI-assisted inventorship.” In other words, a human as the inventor who uses AI as a tool for achieving a specific result (Almarzoqi & Albakjaji, 2022). Third, individual jurisdictions are inconsistent. For example, South Africa issued a patent with DABUS as the inventor, while the EPO, USPTO, and the majority of other patent offices have refused to follow the lead of South Africa. Thus, forum shopping becomes a critical concern because patentees will be inclined to seek more generous jurisdictions to obtain rights that other countries would not grant (Çelik, 2024).

The most apparent difference between the approaches of WIPO, the EPO, and Uzbekistan to AI-generated invention lies in their interpretations of the definition of an inventor. The treaties administered by WIPO and the Paris Convention and the TRIPS Agreement represent the perspective that considers an inventor to be only a natural person. This perspective reflects the 19th- and 20th-century paradigm that informs modern copyright and patent law (Stanková, 2021). Therefore, WIPO’s position reflects existing practice and trends in patent law. The EPO perspective is the most detailed on AI inventorship, which is discussed in the following section.

Legal reasoning behind EPO and WIPO in relation to AI inventorship

The DABUS case heard by the EPO in 2019 involved more than the specific dispute details (Pavlyuk et al., 2021). Thus, machine inventorship was formally incompatible with the EPC unless the European Patent Convention was revised accordingly (Lada, 2023). The EPO rejected the applications, arguing that only a natural person could be appointed inventor under Article 81 of the EPC. This is because legal personality is required to hold rights and duties, which AI cannot (Chesterman, 2020). This way, the EPO confirmed that the central pillar of the European patent law system is the human factor, particularly a living person who is accountable and authorised to perform actions such as inventing. The Legal Board of Appeal explicitly confirmed the incompatibility between the EPC and the notion of an AI system as an inventor in the decisions J8/20 and J9/20. It found that “an inventor must have legal capacity” (Wihlsson, 2022). The EPO’s reasoning behind these legal decisions was that AI lacked intent, comprehension, and conceptualisation, which are integral components of the EPO’s understanding of inventing as a uniquely human endeavour (Kim, 2025); (Resnik & Hosseini, 2025).

Academic and legal debates about the DABUS decision have continued for years, with the central question being whether the EPO's decision is formally correct but substantively wrong. Proponents of expanding current interpretations of inventorship to include AI argue that if an invention by an AI system meets the requirements of novelty and inventiveness, its inventorship by a human inventor is merely a pretextual difference in legal formalism (Nguyen & Quan, 2023). However, the opposite view argues that inventing is a social and legal process whose purpose is to guarantee accountability (Brown, 2021). Thus, by recognising AI systems as inventors or granting them rights traditionally reserved for inventors, the EPO may fail its primary mission of ensuring legal accountability (Marshall, 2023).

With the EPO decisions now officially clarifying that inventorship is a human activity, the practice of "AI-assisted inventions" allows the patenting of AI-generated inventions (Duque Lizarralde & Contreras, 2022). In fact, the EPO's "contributory inventorship" (WIPO, 2022) concept is similar to WIPO's notion of inventorship and legal liability, particularly regarding the central role of humans in inventing and owning intellectual property. Therefore, the EPO decisions implicitly allow for AI-assisted inventions without formally granting inventorship to an AI system.

WIPO's legal reasoning plays a somewhat different role compared to the EPO's explicit statements. As an intergovernmental organisation, WIPO does not review patent applications, and the WIPO Conversations on IP and AI mainly take the form of policy statements (Odita, 2024). Therefore, WIPO explicitly opposes AI inventorship, mainly because it contradicts the treaty obligations to which WIPO members adhere. In other words, WIPO views AI inventorship as a harmonisation issue and opposes it not on theoretical grounds, but because of the practical challenges that divergence from existing treaties may entail (Y. Ebrahim, 2020). While WIPO's statements are less explicit than the EPO's official decisions, they are conceptually similar in their opposition to AI inventorship and their recognition of its implications. Under the current legal framework, AI systems cannot be regarded as legal inventors.

AI, Green Patent, and Technology Transfer

The disclosure-based accountability model proposed above bears directly on environmental law, since a growing share of AI-assisted inventions concern climate-mitigation and green technologies, from battery chemistries to carbon-capture catalysts. The EPO already tags such inventions under its Y02/Y04S classification scheme, and WIPO's GREEN database facilitates transfer of environmentally sound technologies under Article 10 of the Paris Agreement and Article 66 of TRIPS, provisions obliging developed states to support technology transfer to countries such as Uzbekistan. Requiring AI-disclosure for green-technology applications would let Uzbekistan's IP Department identify and prioritise such inventions consistently with its Green Economy Strategy 2019-2030 (Presidential Resolution PP-4477, 2019), linking patent-law harmonisation to Uzbekistan's environmental-law obligations.

This environmental dimension sharpens the attribution-injustice concern identified above: because the Paris Agreement's technology-transfer provisions and Uzbekistan's Green Economy Strategy depend on timely diffusion of climate technologies, unresolved AI-inventorship rules risk delaying the very transfer mechanisms meant to serve those obligations, since technology partners typically condition transfer on clear evidence of who holds enforceable rights. Disclosure of AI involvement therefore serves both the accountability rationale developed above and Uzbekistan's reporting needs under WIPO GREEN.

AI Inventorship Recognition under Uzbekistan's Framework

Uzbekistan's legal framework does not currently recognise AI inventorship. Three distinct pieces of evidence suggest this conclusion. First, the inventor is defined as "a natural person whose creative work creates the invention" in Article 6 of the Law on Inventions, Utility Models and Industrial Designs, which closely resembles Article 81 of the EPC. Second, the recently adopted Law on AI defines AI for informational governance purposes but does not mention patent doctrines or entitlement to protection for AI-generated inventive outputs. Lastly, Uzbekistan's obligations under the Patent Cooperation Treaty also prevent it from formally recognising AI inventorship. According to PCT Rule 19, only natural persons can be designated as inventors (Whitlow, 2020).

Uzbekistan can address this gap by introducing disclosure procedures for AI's involvement in the patent-creation process, providing specific examiner guidance for AI-related applications, and ensuring the technical contribution is attributed to the relevant human agent (Robinson, 2026). While Uzbekistan has yet to recognise AGIs as objects of IP, its legal and institutional infrastructure is steadily adapting to AI innovation, reflecting a proactive, incrementalist regulatory approach that balances opportunity against liability (Zaidan & Ibrahim, 2024). Rather than attempting to redefine inventorship, Uzbekistan should adopt disclosure rules similar to the EPO's, since the harmonisation gap across all three jurisdictions concerns institutional adaptation rather than the definition of inventorship itself.

The first issue is primarily normative. AI regulations include a definition of AI limited to information governance purposes. The definition is entirely dissociated from patent doctrines, meaning patent law is not adjusted to accommodate AI-specific considerations. The normative gap between Uzbekistan's AI policy and patent practice is structural, as the former provides no guidance for the latter. Second, a distinct institutional problem arises from the IP Department's lack of specific guidance on processing AI-related patent applications. The EPO adopted AI Examination Guidelines in 2022, whereas Uzbekistan has yet to introduce any similar IP Department recommendations or guidelines for patent examiners (Chesterman, 2025). As a result, the potential for innovation and commercialisation of AI advancements is

significantly limited, as these factors collectively undermine the national innovation ecosystem.

Bundle theory views legal personhood as a set of distinct legal incidents, including rights, duties, standing, and liability. Using the bundle theory to analyse the prospect of AI inventorship provides greater analytical flexibility in understanding how and why different countries have adopted divergent approaches to the issue (Hattori et al., 2024). In practice, this would involve redistributing some incidents in the bundle of legal personhood to ensure that AI systems, which possess no single incident in the bundle theory, are not granted inventorship (Koulu, 2020). Specifically, one can argue that inventorship requires at least three incidents, namely the “capacity to hold rights, the capacity to bear liability, and the standing to sue and be sued” (S. Afshar, 2022).

Traditional legal liability doctrines and principles apply to human beings only, who are capable of making decisions and taking action, which is not the case with AI systems. Notably, AI systems perform their functions via layers of algorithms, incorporating training data which are unpredictable and unmanageable at multiple levels (Papagiannidis et al., 2023). As a result, responsibility for the actions of AI systems cannot be determined on a case-by-case basis. Patent law doctrines have recognised this principle, which is why AI inventorship remains highly contentious. In particular, the EU AI Liability Directive 2024 contains specific provisions pertaining to causation, thereby creating a rebuttable presumption of causation for certain AI-related harms (Noto La Diega & Bezerra, 2024). This approach demonstrates how the EU is committed to maintaining the principle of human accountability for AI-related damages. The same principle can apply to regulating AI inventorship.

The EU AI Act provides another useful example, showing how accountability can be enhanced by integrating AI regulations with patent law (Söderlund, 2025). This framework does not impose direct obligations on patent law; nevertheless, by requiring developers to conduct extensive documentation and transparency assessments, the AI Act reinforces the EPO’s accountability-focused approach to AI inventorship. In particular, implementing this approach in Uzbekistan would enable the adoption of AI disclosure rules requiring applicants to identify the AI system used to generate the inventive concept and the nature of its contribution to the inventive process (Shen et al., 2026).

The EU regulatory strategy concerning AI is essentially human-centred, emphasising the role of AI technologies in supporting and complementing human capabilities as well as avoiding creating autonomous AI systems capable of exercising normativity (Presno Linera & Meuwese, 2025). For instance, the EPO has adopted AI-specific disclosure requirements for patent applications, which enhance the accountability of human applicants for the actions of AI systems used in the inventive process. One of the EU’s ethical principles suggests that AI systems should be transparent and explainable, which is particularly important for regulating AI

inventorship. Under this principle, any AI-generated inventive concept must undergo human review and oversight to ensure ethical compliance. WIPO's position on the regulation of AI inventorship is similarly grounded in ethical principles, which focus on preserving the integrity of the IP system as well as ensuring that recognition of AI's contribution to innovation does not detract from the role of human creators within the international legal framework (Picht & Thouvenin, 2023). Under this approach, the explainability of AI systems is also a critical requirement that must be imposed on patent applicants using such systems to generate inventive concepts (Rosati, 2025).

Ethical considerations play an increasingly important role in Uzbekistan's AI governance strategy, which is currently being developed. While Uzbekistan's AI Development Strategy until 2030 primarily promotes the adoption of AI technologies to facilitate the country's economic modernisation and technological development, recent regulatory trends suggest that Uzbekistan is beginning to embrace international ethical norms and principles. Notably, the ethical guidelines issued by UNESCO, the OECD, and WIPO, which serve as global frameworks for AI governance, promote a human-centred approach to AI development that aligns with the EPO and WIPO's approaches to regulating AI inventorship (Perboli et al., 2025). In particular, applying these principles to patent law would enhance the accountability of human creators of AI systems, helping balance innovation with the rights of inventors and patent applicants (Gaffar & Albarashdi, 2025).

All three approaches to the regulation of AI inventorship address the central question of whether AI systems should be considered legal persons, but each takes a different approach to the issue. In particular, while the EPO approach focuses on ensuring that legal accountability remains with humans, the WIPO approach examines inventorship in the context of international obligations and legal harmonisation. At the same time, Uzbekistan's approach to the regulation of AI inventorship is complicated by the absence of specific legislation, which creates additional challenges in terms of balancing the need to facilitate AI innovation while protecting the rights of inventors and patent applicants.

Harmonising Uzbekistan's Framework with WIPO and EPO Standards

A comparative analysis of WIPO, the EPO, and Uzbekistan reveals three areas of variance. On definition, all three apply a natural-person rule: WIPO considers an inventor a person under the relevant treaties, the EPO refers to Article 81 EPC and the DABUS jurisprudence, and Uzbekistan adheres to Article 6 of its Patent Law (Benoliel et al., 2024). In particular, Uzbekistan could adopt a statutory disclosure requirement, whereby patent applicants seeking to protect AI-developed innovations would be obliged to identify (a) the AI used, (b) its contribution to the inventive step, and (c) the human input/influence. This obligation would build on the EPO practice and the transparency emerging from WIPO's policy advisory work. According to the IP literature, balancing the desirability of protecting innovators with the need to ensure

adequate access to knowledge requires both parties to contribute to institutional design (AllahRakha, 2025b).

Table.1 Comparative Analysis

Dimension	WIPO	EPO	Uzbekistan
Definition of inventor	Natural person (treaty interpretation)	Natural person (Art. 81 EPC)	Natural person (Art. 6, Law on Inventions 2002)
Basis for natural-person rule	TRIPS & Paris Convention	DABUS decisions J8/20, J9/20	Statutory text; no case law yet
AI-assisted inventions	Accommodated under “AI-assisted inventorship” framework	Permitted since 2022 with human inventor named	No specific provision
Disclosure of AI involvement	Recommended in policy discussions	Required in examination practice	No requirement
Examination guidelines for AI	Policy recommendations only	Dedicated AI Examination Guidelines (2022)	None issued
Accountability mechanisms	Coordination through policy dialogue	AI Liability Directive (proposed); EPO transparency rules	General civil law only; no AI-specific rules
Procedural recognition of AI contribution	Recommended in policy	Recognised via AI Examination Guidelines	Not yet established

Source: Authors, 2026

Analysis of Southeast Asian jurisdictions reveals critical similarities between Uzbekistan and this region in legal responses to the challenges of AI inventorship. In particular, a comparative analysis of the patent legislation and policy of Singapore, Malaysia, Vietnam, Thailand, the Philippines, and Uzbekistan shows that despite active development of AI strategies, none of these jurisdictions has adopted measures to amend the inventorship rules (Oluwatomisin Lamina, 2024). Notably, Singapore, the most developed economy in the region, has adopted a human-centric approach aligned with the WIPO framework. In particular, the Intellectual Property Office of Singapore (IPOS) has issued guidelines that confirm that only natural persons may be registered as inventors. At the same time, IPOS has advocated for disclosing the use of AI in the development of inventions. The EPO has also endorsed this disclosure-without-

inventorship model. Other Southeast Asian jurisdictions, including Malaysia, Vietnam, Thailand, the Philippines, and Uzbekistan, have similarly retained the natural-person rule, with no changes to their patent legislation.

Table. 2 AI Inventorship Position in Southeast Asian Jurisdictions

Jurisdiction	Inventorship Rule	Guidelines	AI Strategy	Alignment
Singapore	Natural person (IPOS guidelines)	Disclosure encouraged; no AI inventor	Singapore’s National AI Strategy 2.0	High: closest to EPO disclosure practice
Malaysia	Natural person (Patents Act 1983)	None issued	Malaysia AI Roadmap 2021–2025	Partial: no procedural instruments
Vietnam	Natural person (IP Law 50/2005/QH11)	None issued	National Strategy for AI (2021)	Low: gap comparable to Uzbekistan
Philippines	Natural person (RA 8293-IP Code 1997)	None issued	Philippine AI Roadmap	Low: no AI-patent interface
Thailand	Natural person (Patent Act B.E. 2522-1979)	None issued	AI Ethics Guideline (2025)	Low: statutory text untested
Uzbekistan	Natural person only	None issued	AI Development Strategy 2030	Partial: doctrinal alignment, no procedures

Source: Authors, 2026

Table 2 demonstrates that Uzbekistan’s rule and procedural gaps are generally comparable to the rules and practice of the Southeast Asian jurisdictions, with the exception of Singapore, which has adopted the most progressive position (Thaldar & Naidoo, 2021).

Five recommendations for Uzbekistan emerge, encompassing (1) mandatory disclosure of AI use in invention development, consistent with EPO practice (Aboy et al., 2024); (2) adoption of EPO-style AI Examination Guidelines 2022 (Yanisky-Ravid & Jin, 2020); (3) coordination with WIPO and EPO, including joint examiner training (Wang & Maynard, 2025); (4) regulatory sandbox pilot projects (Guio Español &

Koenig, 2025); and (5) integration of UNESCO/OECD ethical principles. None requires legislative change to Uzbekistan's Patent Law (Mochizuki et al., 2025).

Uzbekistan is a member of the Eurasian Patent Organization (EAPO), which was established to facilitate the harmonisation and simplification of the procedures for protecting intellectual property in the former Soviet Union. The EAPO member states include Armenia, Belarus, Kazakhstan, Kyrgyzstan, Russia, Tajikistan, Turkmenistan, Uzbekistan, and Azerbaijan. In particular, the EAPO has begun developing harmonised examination practices for handling applications for AI-developed inventions in coordination with WIPO. It is therefore logical that Uzbekistan explore opportunities to leverage the EAPO platform to align its practices with those of the EPO and WIPO, while minimising the transaction costs of direct coordination with the EPO (Straus, 2021).

Examination guidelines issued by the EAPO would automatically apply to all Eurasian patents, including those designated in Uzbekistan (Straňák, 2026). Prior research on Uzbekistan's patent law reforms has not considered this opportunity. Three philosophical dimensions of the natural-person doctrine could help explain why WIPO, the EPO, and Uzbekistan have retained this rule (Igbokwe, 2024). An inventor's rights and obligations should be commensurate with their ability to exercise reason and be held accountable for their actions (Ramos-Zaga, 2025).

These principles helped WIPO, the EPO, and Uzbekistan to respond to the challenges of AI inventorship. Under the natural-person doctrine, an AI-developed invention could be registered only if a natural person had fulfilled the inventorship requirements. While this approach does not answer whether an AI system should be considered an inventor, it addresses more pressing practical concerns, including the need to harmonise practices across jurisdictions. This philosophical position has enabled these jurisdictions to adopt a principled but pragmatic approach to the regulation of AI inventorship

However, this position raises several normative questions about AGIs: 1) Do such systems have the legal capacity to be inventors? 2) Do they have the legal capacity to be sued as inventors? 3) Is it just to hold such systems accountable for their actions, given that they are not capable of reasoning or making choices? A justice-based theory of patent law should therefore stipulate that mandatory disclosure, mediated attribution, and proportional liability would enable AGIs to be held accountable while balancing stakeholders' interests (Poddar & Rao, 2024).

Hybrid Legal Model for Uzbekistan Based on WIPO and EPO Practice

This theoretical nuance is especially pertinent to the Uzbek context, as it allows the country to fulfil its due diligence obligations vis-à-vis global trends in patent law while avoiding the need to alter its national legislation to accommodate the specifics of AI-driven innovation (Wang, 2025). The hybrid model, developed on the basis of WIPO's and the EPO's practices, applies WIPO's coordinating role in determining the

status of the human inventor and the EPO's accountability procedures, including mandatory disclosure and specialised examiner training. Notably, the hybrid model does not entail any legal amendments on the part of Uzbekistan, as it merely builds on the existing practices of the EAPO (Zhang, 2025).

The debate over the legal status of inventors in the context of AI traditionally focused on a binary opposition between recognising AI as an inventor and denying such recognition. The hybrid model transcends that binary opposition by providing a basis for procedural recognition of AI inventorship. This approach is grounded in ontological and axiological analysis showing that inventorship, like authorship, is a relational concept that emerges from the accountability framework of patent law (Ding & Deng, 2025). Thus, patent law is not obliged to grant AI legal personhood, but it may adapt existing procedural accountability mechanisms to incorporate AI's role in the inventive process. In the context of Uzbekistan, such an approach would enable the EAPO to harmonise its practices with those of WIPO, as mandatory AI disclosure and specialised examiner training can be implemented without modifying Uzbek national legislation.

CONCLUSION

This study examines WIPO and EPO responses to AI inventorship and assesses whether Uzbekistan's law could accommodate similar approaches. However, while the EPO has already developed a procedural framework for examining patents involving AI, WIPO adopted a conceptual policy on harmonising inventorship in cases of AI-generated inventions, and Uzbekistan has no specific rules on the subject. However, this provision, which states that only natural persons can be inventors, aligns with the EPO's natural-person requirement. Therefore, this research finds that amending Article 6 of the Law on Inventions is unnecessary. Instead, three measures can help Uzbekistan align its patent legislation with that of WIPO and the EPO. First, patent applicants are required to disclose the use of AI in patent development. Second, Uzbekistan should adopt additional guidelines for examining such patent applications. Finally, it is necessary to cooperate with WIPO, the EPO, and the EAPO on the harmonisation of regulations. These same mechanisms would let Uzbekistan align its AI-patent reforms with its environmental-law commitments under the Paris Agreement: tagging AI-assisted green-technology applications under the EPO's Y02/Y04S scheme would let the IP Department satisfy patent-disclosure and climate-technology-transfer obligations simultaneously, without amending the Law on Inventions.

ACKNOWLEDGMENTS

The authors gratefully acknowledge the guidance of their advisor and the valuable assistance of all individuals who contributed to the completion of this study.

REFERENCES

- Abbott, R. (2020). *The Reasonable Robot: Artificial Intelligence and the Law* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781108631761>
- Aboy, M., Lath, A., Minssen, T., & Liddell, K. (2024). The sufficiency of disclosure of AI inventions. *Journal of Intellectual Property Law and Practice*, 19(11), 834–840. <https://doi.org/10.1093/jiplp/jpac063>
- Aboy, M., Liddell, K., & Lath, A. (2025). Inventorship in the age of AI: Examining the USPTO Guidance on AI-assisted inventions. *Journal of Intellectual Property Law and Practice*, 20(7), 495–502. <https://doi.org/10.1093/jiplp/jpaf019>
- Ahmed Mohamed, N., Mohamad Salem Rahma, M., & Al Hemairy, M. (2025). Challenges and prospects in granting patent rights to AI-generated inventions: A global legal study with special reference to the UAE. *Humanities and Social Sciences Communications*, 12(1), 1851. <https://doi.org/10.1057/s41599-025-06121-y>
- Al-Busaidi, A. S., Raman, R., Hughes, L., Albashrawi, M. A., Malik, T., Dwivedi, Y. K., Al- Alawi, T., AlRizeiqi, M., Davies, G., Fenwick, M., Gupta, P., Gurple, S., Hooda, A., Jurcys, P., Lim, D., Lucchi, N., Misra, T., Raman, R., Shirish, A., & Walton, P. (2024). Redefining boundaries in innovation and knowledge domains: Investigating the impact of generative artificial intelligence on copyright and intellectual property rights. *Journal of Innovation & Knowledge*, 9(4), 100630. <https://doi.org/10.1016/j.jik.2024.100630>
- AllahRakha, N. (2025a). Legislators Qualifications in Pakistan Under Islamic Constitutional Provisions. *Journal of Human Rights, Culture and Legal System*, 5(2), 473–499. <https://doi.org/10.53955/jhcls.v5i2.491>
- AllahRakha, N. (2025b). The Legality Of Reverse Engineering And The Protection Of Trade Secrets In The Software Industry. *Jurisdictie: Jurnal Hukum Dan Syariah*, 15(2), 309–336. <https://doi.org/10.18860/j.v15i2.28422>
- Almarzoqi, R., & Albakjaji, M. (2022). The Patentability of AI Invention: The Case of the Kingdom of Saudi Arabia Law. *International Journal of Service Science, Management, Engineering, and Technology*, 13(1), 1–22. <https://doi.org/10.4018/IJSSMET.307111>
- Balganesh, S., Sichelman, T. M., & Smith, H. E. (Eds.). (2022). *Wesley Hohfeld A Century Later: Edited Work, Select Personal Papers, and Original Commentaries* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781108131742>

- Bayındır, A. S., & Danaher, J. (2025). Why We Should Recognize AI as an Inventor. *Journal of Bioethical Inquiry*, 22(3), 515–525. <https://doi.org/10.1007/s11673-025-10429-4>
- Benoliel, D., Yu, P. K., Gurry, F., & Lee, K. (Eds.). (2024). Intellectual Property and Global Inequality: In *Intellectual Property, Innovation and Economic Inequality* (1st ed., pp. 277–278). Cambridge University Press. <https://doi.org/10.1017/9781108894722.014>
- Błaszczuk, M. (2024). Impossibility of Artificial Inventors. *University of Michigan Law School*, 16(8), 73–96.
- Brown, R. D. (2021). Property ownership and the legal personhood of artificial intelligence. *Information & Communications Technology Law*, 30(2), 208–234. <https://doi.org/10.1080/13600834.2020.1861714>
- Çelik, F. B. (2024). Virtues and Perils of Forum-Shopping in European Security. *European Security*, 33(1), 1–20. <https://doi.org/10.1080/09662839.2023.2222666>
- Chesterman, S. (2020). Artificial Intelligence And The Limits Of Legal Personality. *International and Comparative Law Quarterly*, 69(4), 819–844. <https://doi.org/10.1017/S0020589320000366>
- Chesterman, S. (2025). Good models borrow, great models steal: Intellectual property rights and generative AI. *Policy and Society*, 44(1), 23–37. <https://doi.org/10.1093/polsoc/puae006>
- DiMatteo, L. A., Poncibò, C., & Cannarsa, M. (Eds.). (2022). *The Cambridge Handbook of Artificial Intelligence: Global Perspectives on Law and Ethics* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781009072168>
- Ding, W., & Deng, S. (2025). The Patentability of AI-Generated Technical Solutions and Institutional Responses: Chinese Perspective vs. Other Countries. *Information*, 16(8), 629. <https://doi.org/10.3390/info16080629>
- Duque Lizarralde, M., & Contreras, H. A. (2022). The real role of AI in patent law debates. *International Journal of Law and Information Technology*, 30(1), 23–46. <https://doi.org/10.1093/ijlit/eaac008>
- Gaffar, H., & Albarashdi, S. (2025). Copyright Protection for AI-Generated Works: Exploring Originality and Ownership in a Digital Landscape. *Asian Journal of International Law*, 15(1), 23–46. <https://doi.org/10.1017/S2044251323000735>
- Guio Español, A., & Koenig, P. D. (2025). Regulatory sandboxes for AI in the majority world: A learning-centric approach to legal adaptation. *Cambridge Forum on AI: Law and Governance*, 1, e42. <https://doi.org/10.1017/cfl.2025.10015>
- Hattori, E. A., Yamakawa, M., & Miwa, K. (2024). Human bias in evaluating AI product creativity. *Journal of Creativity*, 34(2), 100087. <https://doi.org/10.1016/j.yjoc.2024.100087>

- Igbokwe, E. M. (2024). Human to machine innovation: Does legal personhood and inventorship threshold offer any leeway? *The Journal of World Intellectual Property*, 27(2), 149–174. <https://doi.org/10.1111/jwip.12294>
- Kim, D. (2020). ‘AI-Generated Inventions’: Time to Get the Record Straight? *GRUR International*, 69(5), 443–456. <https://doi.org/10.1093/grurint/ikaa061>
- Kim, D. (2025). The Illusory Standard of Significant Human Contribution to AI-Assisted Inventions after the DABUS Decision of the German Federal Court of Justice. *IIC - International Review of Intellectual Property and Competition Law*, 56(2), 369–380. <https://doi.org/10.1007/s40319-025-01567-8>
- Kim, D., Alber, M., Kwok, M. W., Mitrović, J., Ramirez-Atencia, C., Pérez, J. A. R., & Zille, H. (2022). Clarifying Assumptions About Artificial Intelligence Before Revolutionising Patent Law. *GRUR International*, 71(4), 295–321. <https://doi.org/10.1093/grurint/ikab174>
- Koulu, R. (2020). Proceduralizing control and discretion: Human oversight in artificial intelligence policy. *Maastricht Journal of European and Comparative Law*, 27(6), 720–735. <https://doi.org/10.1177/1023263X20978649>
- Kurki, V. A. J. (2019). The Incidents of Legal Personhood. In V. A. Kurki, *A Theory of Legal Personhood* (1st ed., pp. 91–126). Oxford University Press Oxford. <https://doi.org/10.1093/oso/9780198844037.003.0004>
- Kwakwa, E. (2019). Intellectual Property. In S. Chesterman, D. M. Malone, & S. Villalpando (Eds.), *The Oxford Handbook of United Nations Treaties* (1st ed., pp. 363–374). Oxford University Press. <https://doi.org/10.1093/law/9780190947842.003.0022>
- Lada, M. (2023). Artificial intelligence, inventorship and the myth of the inventing machine: Can a process be an inventor? *Information & Communications Technology Law*, 32(3), 288–327. <https://doi.org/10.1080/13600834.2022.2154049>
- Li, W., Song, R., Zhang, B., & Yu, K. (2025). AI creativity and legal protection for AI-generated works in posthuman societal scenarios. *Sustainable Futures*, 9, 100749. <https://doi.org/10.1016/j.sftr.2025.100749>
- Mahlmann, M. (2023). *Mind and Rights: The History, Ethics, Law and Psychology of Human Rights* (1st ed.). Cambridge University Press. <https://doi.org/10.1017/9781316875520>
- Marshall, B. (2023). No legal personhood for AI. *Patterns*, 4(11), 100861. <https://doi.org/10.1016/j.patter.2023.100861>
- Matsieli, M., & Mutula, S. (2025). Generative AI and the Information Society: Ethical Reflections from Libraries. *Information*, 16(9), 771. <https://doi.org/10.3390/info16090771>
- Matulionyte, R. (2025). ‘AI is not an Inventor’: *Thaler v Comptroller of Patents, Designs and Trademarks* and the Patentability of AI Inventions. *The Modern Law Review*, 88(1), 205–218. <https://doi.org/10.1111/1468-2230.12907>

- Mochizuki, Y., Bruillard, E., & Bryan, A. (2025). The ethics of AI or technosolutionism? UNESCO's policy guidance on AI in education. *British Journal of Sociology of Education*, 1–22. <https://doi.org/10.1080/01425692.2025.2502808>
- Nguyen, N. K. H., & Quan, D. H. (2023). Artificial Intelligence and Inventorship Under the Patent Law Regime: Practical Development from Common Law Jurisdictions. *Vietnamese Journal of Legal Sciences*, 8(1), 25–54. <https://doi.org/10.2478/vjls-2023-0002>
- Noto La Diega, G., & Bezerra, L. C. T. (2024). Can there be responsible AI without AI liability? Incentivizing generative AI safety through ex-post tort liability under the EU AI liability directive. *International Journal of Law and Information Technology*, 32, eaae021. <https://doi.org/10.1093/ijlit/eaae021>
- Novelli, C., Casolari, F., Hacker, P., Spedicato, G., & Floridi, L. (2024). Generative AI in EU law: Liability, privacy, intellectual property, and cybersecurity. *Computer Law & Security Review*, 55, 106066. <https://doi.org/10.1016/j.clsr.2024.106066>
- Odita, O. (2024). The role of WIPO in regulating AI and IP at the global level. *GSC Advanced Research and Reviews*, 21(3), 470–476. <https://doi.org/10.30574/gscarr.2024.21.3.0504>
- Oluwatomisin Lamina. (2024). *Generative Ai, Inventorship And The Future Of Patent Policy*. <https://doi.org/10.13140/RG.2.2.33917.27365>
- Oriakhogba, D. O. (2021). What If DABUS Came to Africa? Visiting AI Inventorship and Ownership of Patent from the Nigerian Perspective. *Business Law Review*, 42(Issue 2), 89–99. <https://doi.org/10.54648/BULA2021013>
- Papagiannidis, E., Mikalef, P., Conboy, K., & Van De Wetering, R. (2023). Uncovering the dark side of AI-based decision-making: A case study in a B2B context. *Industrial Marketing Management*, 115, 253–265. <https://doi.org/10.1016/j.indmarman.2023.10.003>
- Perboli, G., Simionato, N., & Pratali, S. (2025). Navigating the AI regulatory landscape: Balancing innovation, ethics, and global governance. *Economic and Political Studies*, 13(4), 367–397. <https://doi.org/10.1080/20954816.2025.2569584>
- Picht, P. G., & Thouvenin, F. (2023). AI and IP: Theory to Policy and Back Again – Policy and Research Recommendations at the Intersection of Artificial Intelligence and Intellectual Property. *IIC - International Review of Intellectual Property and Competition Law*, 54(6), 916–940. <https://doi.org/10.1007/s40319-023-01344-5>
- Poddar, A., & Rao, S. R. (2024). Evolving intellectual property landscape for AI-driven innovations in the biomedical sector: Opportunities in stable IP regime for shared success. *Frontiers in Artificial Intelligence*, 7, 1372161. <https://doi.org/10.3389/frai.2024.1372161>
- Presno Linera, M. Á., & Meuwese, A. (2025). Regulating AI from Europe: A joint analysis of the AI Act and the Framework Convention on AI. *The Theory and*

- Practice of Legislation*, 13(3), 292–311.
<https://doi.org/10.1080/20508840.2025.2492524>
- Ramos-Zaga, F. A. (2025). Reconceptualizing Human Authorship in the Age of Generative AI: A Normative Framework for Copyright Thresholds. *Laws*, 14(6), 84. <https://doi.org/10.3390/laws14060084>
- Resnik, D. B., & Hosseini, M. (2025). The ethics of using artificial intelligence in scientific research: New guidance needed for a new tool. *AI and Ethics*, 5(2), 1499–1521. <https://doi.org/10.1007/s43681-024-00493-8>
- Robinson, W. K. (2026). Responsibility, Transparency, and Accountability in AI Patents. In D. Lim & P. K. Yu (Eds.), *Inclusive Innovation in the Age of AI and Big Data* (1st ed., pp. 279–298). Oxford University Press New York, NY. <https://doi.org/10.1093/9780197799444.003.0018>
- Rodrigues, R. (2020). Legal and human rights issues of AI: Gaps, challenges and vulnerabilities. *Journal of Responsible Technology*, 4, 100005. <https://doi.org/10.1016/j.jrt.2020.100005>
- Rosati, E. (2025). Infringing AI: Liability for AI-Generated Outputs under International, EU, and UK Copyright Law. *European Journal of Risk Regulation*, 16(2), 603–627. <https://doi.org/10.1017/err.2024.72>
- S. Afshar, M. (2022). *Artificial Intelligence and Inventorship – Does the Patent Inventor Have to be Human?* 13(1), 55–72.
- Sabbagh, C. (2001). A Taxonomy of Normative and Empirically Oriented Theories of Distributive Justice. *Social Justice Research*, 14(3), 237–263. <https://doi.org/10.1023/A:1014315923278>
- Sabel Nastasya, V., & Santoso, B. (2025). Analysis of The Swiss Court Decision on The Rejection of AI as An Inventor in International Patent Law. *International Journal of Social Science and Human Research*, 08(09). <https://doi.org/10.47191/ijsshr/v8-i9-101>
- Shen, J. H., Liu, K., Wang, A., Cen, S. H., Zhang, A. K., Meinhardt, C., Zhang, D., Klyman, K., Bommasani, R., & Ho, D. E. (2026). *The Limits of AI Data Transparency Policy: Three Disclosure Fallacies* (arXiv:2601.18127). arXiv. <https://doi.org/10.48550/arXiv.2601.18127>
- Söderlund, K. (2025). High-risk AI transparency? On qualified transparency mandates for oversight bodies under the EU AI Act. *Technology and Regulation*, 2025, 97–113. <https://doi.org/10.71265/6bedar76>
- Stanková, E. (2021). Human Inventorship In European Patent LAW. *The Cambridge Law Journal*, 80(2), 338–365. <https://doi.org/10.1017/S0008197321000507>
- Straňák, P. (2026). What Artificial Intelligence May Be Missing—And Why It Is Unlikely to Attain It Under Current Paradigms. *Philosophies*, 11(1), 20. <https://doi.org/10.3390/philosophies11010020>

- Straus, J. (2021). Will Artificial Intelligence Change Some Patent Law Paradigms? *Ljubljana Law Review*, 81(1), 11–61. <https://doi.org/10.51940/2021.1.11-61>
- Thaldar, D., & Naidoo, M. (2021). AI inventorship: The right decision? *South African Journal of Science*, 117(11/12). <https://doi.org/10.17159/sajs.2021/12509>
- Wang, J. (2025). Navigating the USPTO's AI inventorship guidance in AI-driven drug discovery. *Journal of Law and the Biosciences*, 12(2), lsaf014. <https://doi.org/10.1093/jlb/lsaf014>
- Wang, J., & Maynard, A. (2025). Gender disparity in U.S. patenting. *Humanities and Social Sciences Communications*, 12(1), 1730. <https://doi.org/10.1057/s41599-025-06038-6>
- Whitlow, L. (2020). When the Invented Becomes the Inventor: Can, and Should AI Systems be Granted Inventorship Status for Patent Applications? *Legal Issues in the Digital Age*, 2(2), 3–23. <https://doi.org/10.17323/2713-2749.2020.2.3.23>
- Wihlsson, J. (2022, January 13). *The Legal Board of Appeal of the EPO confirms AI cannot be named as inventor on patent applications in Europe*. <https://www.brann.se/en/2022/01/13/epo-confirms-ai-cannot-be-named-inventor/>
- Y. Ebrahim, T. (2020). Artificial Intelligence Inventions & Patent Disclosure. *PENN State Law Review*, 125(1), 147–221.
- Yanisky-Ravid, S., & Jin, R. (2020). Summoning a New Artificial Intelligence Patent Model: In the Age of Pandemic. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3619069>
- Yohanna, F., & Suleiman, S. I. (2025). The Impact Of Artificial Intelligence On Creativity, Innovation And Intellectual Property Rights Innigeria. *Imam Ja'afar Al-Sadiq University Journal of Legal Studies*, 4(2). <https://doi.org/10.64682/3104-9419.1101>
- Zaidan, E., & Ibrahim, I. A. (2024). AI Governance in a Complex and Rapidly Changing Regulatory Landscape: A Global Perspective. *Humanities and Social Sciences Communications*, 11(1), 1121. <https://doi.org/10.1057/s41599-024-03560-x>
- Zhang, T. (2025). Legal Analysis and Challenges of Artificial Intelligence as an Inventor: A Study Based on UK and US Patent Law. *Lecture Notes in Education Psychology and Public Media*, 96(1), 115–120. <https://doi.org/10.54254/2753-7048/2025.BO24151>